QUASI-COMPACTNESS OF A FIELD POINT IS INDEPENDENT OF THE REPRESENTATIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3388-3398 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 3400-3410, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 36 of queue_stacks-properties.json (pb=469, stmt 3388-3398, proof 3400-3410).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 045F: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3410–3410. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 040W: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3410–3410. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $x \in |\mathcal{X}|$ be a point. The following are equivalent*

(1) *some morphism $\mathrm{Spec}(k) \to \mathcal{X}$ in the equivalence class of x is quasi-compact, and*
(2) *any morphism $\mathrm{Spec}(k) \to \mathcal{X}$ in the equivalence class of x is quasi-compact.*

Proof. Let $\mathrm{Spec}(k) \to \mathcal{X}$ be in the equivalence class of x. Let k'/k be a field extension. Then we have to show that $\mathrm{Spec}(k) \to \mathcal{X}$ is quasi-compact if and only if $\mathrm{Spec}(k') \to \mathcal{X}$ is quasi-compact. This follows from Morphisms of Spaces, Lemma 040W and the principle of Algebraic Stacks, Lemma 045F. □

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.